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跨源合并文献

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结构化表格

01 · LITERATURE RETRIEVAL

1 篇跨源记录已合并

多来源文献检索与去重

统一适配 OpenAlex / Crossref，主记录不绑定单一服务

laser welding porosity

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跨源检索

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DOI

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标题+第一作者

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单源记录

Keyhole Stability and Pore Formation During High-Power Laser Welding

Wang Hao, Elena Smirnova · The International Journal of Advanced Manufacturing Technology · 2026

DOI 10.1007/s00170-026-14242-1

引用 7

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Laser Welding Parameter Optimization and Porosity Control in Q355 Steel

Li Ming, Anna Petrova, Chen Wei · Journal of Materials Processing Technology · 2026

DOI 10.1016/j.jmatprotec.2026.118999

引用 18

Laser Welding Parameter Optimization and Porosity

当前定位 P1

Laser Welding Parameter Optimization and Porosity Control

Abstract

This study investigates laser welding of Q355 structural steel and connects processing parameters to mechanical performance and porosity control.

Introduction

Laser welding quality depends on heat input, shielding gas and keyhole stability.

Materials and Methods

The base material was Q355 steel with a thickness of 3 mm. The laser power was 2.4 kW and travel speed was 12 mm/s.

Table 1 Welding parameters

Group	Laser power	Speed
A	2.0 kW	10 mm/s
B	2.4 kW	12 mm/s
C	2.8 kW	14 mm/s

STRUCTURED READING

章节与关键事实

Front Matter · P1

Abstract · P1

Introduction · P1

Materials and Methods · P1

Results · P2

Discussion · P2

Conclusions · P2

References · P2

laser power

2.4 kW

P1 · laser power was 2.4 kW

laser power

2.4 kW

P2 · laser power of 2.4 kW

tensile strength

612 MPa

P2 · tensile strength was 612 MPa

travel speed

12 mm/s

P1 · travel speed was 12 mm/s

PAGE EVIDENCE

文本层解析

原文证据

P1 Materials and Methods

The base material was Q355 steel with a thickness of 3 mm. The laser power was 2.4 kW and travel speed was 12 mm/s.

P2 Results / Discussion

The highest tensile strength was 612 MPa for parameter group B. Porosity formed because shielding gas was trapped during unstable keyhole collapse.



表格数据提取

表格编号、物理页码与结构化二维数据同时保留

04 • TABLE EXTRACTION

结构化成功

Table 1 • Welding parameters

原文第 1 页 • pdfplumber 文本/网格抽取

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```

Table Evidence

P1

可点击回到原文物理页

Rows

3

参数组 A / B / C

Columns

3

Group / Laser power / Speed

Key parameter

2.4 kW

Group B • 12 mm/s



基于论文的原文问答

答案只来自已解析证据，并返回页码、章节、片段和相关度

Evidence-first QA

05 • PAPER QA

问题

激光功率是多少？

high confidence

论文原文给出的激光功率为 2.4 kW。

最高抗拉强度对应哪组参数？

论文报告的最高抗拉强度为 612 MPa，对应参数组 B。

这篇论文使用了什么母材？

论文原文描述的母材为：Q355 steel with a thickness of 3 mm。

作者如何解释气孔形成？

作者将气孔形成解释为：不稳定小孔塌陷过程中保护气体被困（shielding gas was trapped）；稳定的小孔行为可减少气孔形成。

CLICKABLE EVIDENCE

物理页码定位

证据卡片

P1 Materials and Methods

score 0.4518

The laser power was 2.4 kW and travel speed was 12 mm/s.

P2 Conclusions

score 0.3299

Conclusions A laser power of 2.4 kW produced the best combined mechanical performance in this experiment.

P2 Results

score 0.6164

Results The highest tensile strength was 612 MPa for parameter group B.

P2 Discussion

score 0.4614

Discussion Porosity formed because shielding gas was trapped during unstable keyhole collapse; stable keyhole behavior reduced pore formation.



原文页码定位

点击问答或结构化事实中的 P2 证据后，PDF 视图定位到对应物理页

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PDF 原文

P2 / 2

Results

The highest tensile strength was 612 MPa for parameter group B.

Discussion

Porosity formed because shielding gas was trapped during unstable keyhole collapse; stable keyhole behavior reduced pore formation.

Conclusions

A laser power of 2.4 kW produced the best combined mechanical performance in this experiment.

References

- [1] Zhang et al. Laser welding process monitoring. 2024.
- [2] Wang et al. Keyhole stability and pore formation. 2023.

PAGE LOCATOR

P2

定位成功

来自“最高抗拉强度”和“气孔形成原因”的证据均落在 PDF 物理第 2 页。

P2 Results score 0.6164

Results The highest tensile strength was 612 MPa for parameter group B.

P2 Discussion score 0.4614

Discussion Porosity formed because shielding gas was trapped during unstable keyhole collapse; stable keyhole behavior reduced pore formation.

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